Algoma UNIVERSITY

School of Computer Science & Technology - Algoma

Linear Algebra

Math 1057

Instructor

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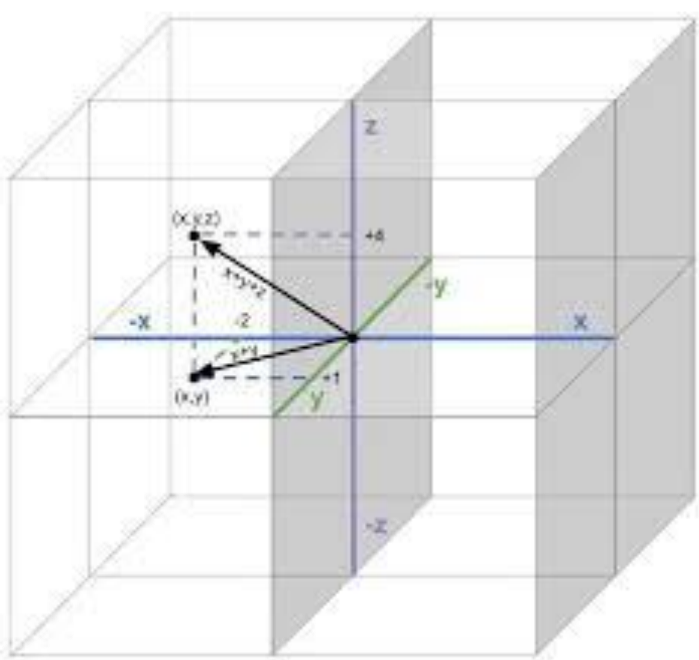

18 Coordinate Systems

Lecture 18

Coordinate Systems

18.1 Coordinates

Recall that a basis of a vector space V is a set of vectors $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in V such that

1. the set $\mathcal{B}$ spans all of V, that is, $\mathsf{V} = \operatorname{span}(\mathcal{B})$, and
2. the set $\mathcal{B}$ is linearly independent.

Hence, if $\mathcal{B}$ is a basis for V, each vector $\mathbf{x}^* \in \mathsf{V}$ can be written as a linear combination of $\mathcal{B}$:

$$\mathbf{x}^* = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

Moreover, from the definition of linear independence given in Definition 6.1, any vector $\mathbf{x} \in \operatorname{span}(\mathcal{B})$ can be written in **only one** way as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$. In other words, for the $\mathbf{x}^*$ above, there does not exist other scalars $t_1, \ldots, t_n$ such that also

$$\mathbf{x}^* = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n.$$

To see this, suppose that we can write $\mathbf{x}^*$ in two different ways using $\mathcal{B}$:

$$\begin{aligned} \mathbf{x}^* &= c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n \\ \mathbf{x}^* &= t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n. \end{aligned}$$

Then

$$\mathbf{0} = \mathbf{x}^* - \mathbf{x}^* = (c_1 - t_1)\mathbf{v}_1 + (c_2 - t_2)\mathbf{v}_2 + \cdots + (c_n - t_n)\mathbf{v}_n.$$

Since $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, the only linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$ that gives the zero vector $\mathbf{0}$ is the trivial linear combination. Therefore, it must be the case that $c_i - t_i = 0$, or equivalently that $c_i = t_i$ for all $i = 1, 2 \ldots, n$. Thus, there is only one way to write $\mathbf{x}^*$ in terms of $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. Hence, relative to the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, the scalars $c_1, c_2, \ldots, c_n$ uniquely determine the vector $\mathbf{x}$, and vice-versa.

Our preceding discussion on the unique representation property of vectors in a given basis leads to the following definition.

Definition 18.1: Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis for V and let $\mathbf{x} \in \mathsf{V}$. The **coordinates of x relative to the basis** $\mathcal{B}$ are the unique scalars $c_1, c_2, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

In vector notation, the $\mathcal{B}$-coordinates of $\mathbf{x}$ will be denoted by

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

and we will call $[\mathbf{x}]_{\mathcal{B}}$ the coordinate vector of $\mathbf{x}$ relative to $\mathcal{B}$.

The notation $[\mathbf{x}]_{\mathcal{B}}$ indicates that these are coordinates of $\mathbf{x}$ with respect to the basis $\mathcal{B}$. If it is clear what basis we are working with, we will omit the subscript $\mathcal{B}$ and simply write $[\mathbf{x}]$ for the coordinates of $\mathbf{x}$ relative to $\mathcal{B}$.

Example 18.2. One can verify that

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^2$. Find the coordinates of $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ relative to $\mathcal{B}$.

Solution. Let $\mathbf{v}_1 = (1, 1)$ and let $\mathbf{v}_2 = (-1, 1)$. By definition, the coordinates of $\mathbf{v}$ with respect to $\mathcal{B}$ are the scalars c_1, c_2 such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

If we put $\mathbf{P} = [\mathbf{v}_1\ \mathbf{v}_2]$, and let $[\mathbf{v}]_{\mathcal{B}} = (c_1, c_2)$, then we need to solve the linear system

$$\mathbf{v} = \mathbf{P}[\mathbf{v}]_{\mathcal{B}}$$

Solving the linear system, one finds that the solution is $[\mathbf{v}]_{\mathcal{B}} = (2, -1)$, and therefore this is the $\mathcal{B}$-coordinate vector of $\mathbf{v}$, or the coordinates of $\mathbf{v}$, relative to $\mathcal{B}$. □

It is clear how the procedure of the previous example can be generalized. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for $\mathbb{R}^n$ and let $\mathbf{v}$ be any vector in $\mathbb{R}^n$. Put $\mathbf{P} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$. Then the $\mathcal{B}$-coordinates of $\mathbf{v}$ is the unique column vector $[\mathbf{v}]_{\mathcal{B}}$ solving the linear system

$$\mathbf{P}\mathbf{x} = \mathbf{v}$$

that is, $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$ is the unique solution to $\mathbf{P}\mathbf{x} = \mathbf{v}$. Because $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, the solution to $\mathbf{P}\mathbf{x} = \mathbf{v}$ is

$$[\mathbf{v}]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{v}.$$

We remark that if an inconsistent row arises when you row reduce the augmented matrix $\begin{bmatrix} \mathbf{P} & \mathbf{v} \end{bmatrix}$ then you have made an error in your row reduction algorithm. In summary, to find coordinates with respect to a basis $\mathcal{B}$ in $\mathbb{R}^n$, we need to solve a square linear system.

Example 18.3. Let

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

and let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$. One can show that $\mathcal{B}$ is linearly independent and therefore a basis for $\mathsf{W} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Determine if $\mathbf{x}$ is in W, and if so, find the coordinate vector of $\mathbf{x}$ relative to $\mathcal{B}$.

Solution. By definition, $\mathbf{x}$ is in $\mathsf{W} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ if we can write $\mathbf{x}$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$$

Form the associated augmented matrix and row reduce:

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

The system is consistent with solution $c_1 = 2$ and $c_2 = 3$. Therefore, $\mathbf{x}$ is in W, and the $\mathcal{B}$-coordinates of $\mathbf{x}$ are

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

□

Example 18.4. What are the coordinates of

$$\mathbf{v} = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix}$$

in the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$?

Solution. Clearly,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 11\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 7\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, the coordinate vector of $\mathbf{v}$ relative to $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is

$$[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 11 \\ -7 \end{bmatrix}$$

$\square$

Example 18.5. Let $\mathbb{P}_3[t]$ be the vector space of polynomials of degree at most 3.

(i) Show that $\mathcal{B} = \{1, t, t^2, t^3\}$ is a basis for $\mathbb{P}_3[t]$.

(ii) Find the coordinates of $v(t) = 3 - t^2 - 7t^3$ relative to $\mathcal{B}$.

Solution. The set $\mathcal{B} = \{1, t, t^2, t^3\}$ is a spanning set for $\mathbb{P}_3[t]$. Indeed, any polynomial $u(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$ is clearly a linear combination of $1, t, t^2, t^3$. Is $\mathcal{B}$ linearly independent? Suppose that there exists scalars c_0, c_1, c_2, c_3 such that

$$c_0 + c_1 t + c_2 t^2 + c_3 t^3 = 0.$$

Since the above equality must hold for all values of t, we conclude that $c_0 = c_1 = c_2 = c_3 = 0$. Therefore, $\mathcal{B}$ is linearly independent, and consequently a basis for $\mathbb{P}_3[t]$. In the basis $\mathcal{B}$, the coordinates of $v(t) = 3 - t^2 - 7t^3$ are

$$[v(t)]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ -7 \end{bmatrix}$$

The basis $\mathcal{B} = \{1, t, t^2, t^3\}$ is called the **standard basis** in $\mathbb{P}_3[t]$. $\square$

Example 18.6. Show that

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is a basis for $M_{2\times 2}$. Find the coordinates of $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ -4 & -1 \end{bmatrix}$ relative to $\mathcal{B}$.

Solution. Any matrix $\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$ can be written as a linear combination of the matrices in $\mathcal{B}$:

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = m_{11} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + m_{12} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + m_{21} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + m_{22} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

If

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

then clearly $c_1 = c_2 = c_3 = c_4 = 0$. Therefore, $\mathcal{B}$ is linearly independent, and consequently a basis for $M_{2\times 2}$. The coordinates of $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ -4 & -1 \end{bmatrix}$ in the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

are

$$[\mathbf{A}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 0 \\ -4 \\ -1 \end{bmatrix}$$

The basis $\mathcal{B}$ above is the **standard basis** of $M_{2\times 2}$. $\square$

18.2 Coordinate Mappings

Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis of $\mathbb{R}^n$ and let $\mathbf{P} = [\mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_n] \in M_{n\times n}$. If $\mathbf{x} \in \mathbb{R}^n$ and $[\mathbf{x}]_{\mathcal{B}}$ are the $\mathcal{B}$-coordinates of $\mathbf{x}$ relative to $\mathcal{B}$ then

$$\mathbf{x} = \mathbf{P}[\mathbf{x}]_{\mathcal{B}}. \tag{$\star$}$$

Hence, thinking of $\mathbf{P} : \mathbb{R}^n \to \mathbb{R}^n$ as a linear mapping, $\mathbf{P}$ maps $\mathcal{B}$-coordinate vectors to coordinate vectors relative to the standard basis of $\mathbb{R}^n$. For this reason, we call $\mathbf{P}$ the **change-of-coordinates matrix** from the basis $\mathcal{B}$ to the standard basis in $\mathbb{R}^n$. If we need to emphasize that $\mathbf{P}$ is constructed from the basis $\mathcal{B}$ we will write $\mathbf{P}_{\mathcal{B}}$ instead of just $\mathbf{P}$. Multiplying equation ($\star$) by $\mathbf{P}^{-1}$ we obtain

$$\mathbf{P}^{-1}\mathbf{x} = [\mathbf{x}]_{\mathcal{B}}.$$

Therefore, $\mathbf{P}^{-1}$ maps coordinate vectors in the standard basis to coordinates relative to $\mathcal{B}$.

Example 18.7. The columns of the matrix $\mathbf{P}$ form a basis $\mathcal{B}$ for $\mathbb{R}^3$:

$$\mathbf{P} = \begin{bmatrix} 1 & 3 & 3 \\ -1 & -4 & -2 \\ 0 & 0 & -1 \end{bmatrix}.$$

(a) What vector $\mathbf{x} \in \mathbb{R}^3$ has $\mathcal{B}$-coordinates $[\mathbf{x}]_{\mathcal{B}} = (1, 0, -1)$.

(b) Find the $\mathcal{B}$-coordinates of $\mathbf{v} = (2, -1, 0)$.

Solution. The matrix $\mathbf{P}$ maps $\mathcal{B}$-coordinates to standard coordinates in $\mathbb{R}^3$. Therefore,

$$\mathbf{x} = \mathbf{P}[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

On the other hand, the inverse matrix $\mathbf{P}^{-1}$ maps standard coordinates in $\mathbb{R}^3$ to $\mathcal{B}$-coordinates. One can verify that

$$\mathbf{P}^{-1} = \begin{bmatrix} 4 & 3 & 6 \\ -1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Therefore, the $\mathcal{B}$ coordinates of $\mathbf{v}$ are

$$[\mathbf{v}]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{v} = \begin{bmatrix} 4 & 3 & 6 \\ -1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}$$

□

When V is an abstract vector space, e.g. $\mathbb{P}_n[t]$ or $M_{n\times n}$, the notion of a coordinate mapping is similar as the case when $\mathsf{V} = \mathbb{R}^n$. If V is an n-dimensional vector space and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for V, we define the coordinate mapping $\mathcal{P} : \mathsf{V} \to \mathbb{R}^n$ relative to $\mathcal{B}$ as the mapping

$$\mathcal{P}(\mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}.$$

Example 18.8. Let $\mathsf{V} = M_{2\times 2}$ and let $\mathcal{B} = \{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_4\}$ be the standard basis for $M_{2\times 2}$. What is $\mathcal{P} : M_{2\times 2} \to \mathbb{R}^4$?

Solution. Recall,

$$\mathcal{B} = \{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_4\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Then for any $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ we have

$$\mathcal{P}\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right) = \begin{bmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{bmatrix}.$$

□

18.3 Matrix Representation of a Linear Map

Let V and W be vector spaces and let $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ be a linear mapping. Then by definition of a linear mapping, $\mathsf{T}(\mathbf{v} + \mathbf{u}) = \mathsf{T}(\mathbf{v}) + \mathsf{T}(\mathbf{u})$ and $\mathsf{T}(\alpha\mathbf{v}) = \alpha\mathsf{T}(\mathbf{v})$ for every $\mathbf{v}, \mathbf{u} \in \mathsf{V}$ and $\alpha \in \mathbb{R}$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis of V and let $\gamma = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ be a basis of W. Then for any $\mathbf{v} \in \mathsf{V}$ there exists scalars $c_1, c_2, \ldots, c_n$ such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

and thus $[\mathbf{v}]_{\mathcal{B}} = (c_1, c_2, \ldots, c_n)$ are the coordinates of $\mathbf{v}$ in the basis $\mathcal{B}$ By linearity of the mapping T we have

$$\begin{aligned}\mathsf{T}(\mathbf{v}) &= \mathsf{T}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n)\\ &= c_1\mathsf{T}(\mathbf{v}_1) + c_2\mathsf{T}(\mathbf{v}_2) + \cdots + c_n\mathsf{T}(\mathbf{v}_n)\end{aligned}$$

Now each vector $\mathsf{T}(\mathbf{v}_j)$ is in W and therefore because γ is a basis of W there are scalars $a_{1,j}, a_{2,j}, \ldots, a_{m,j}$ such that

$$\mathsf{T}(\mathbf{v}_j) = a_{1,j}\mathbf{w}_1 + a_{2,j}\mathbf{w}_2 + \cdots + a_{m,j}\mathbf{w}_m$$

In other words,

$$[\mathsf{T}(v_j)]_\gamma = (a_{1,j}, a_{2,j}, \ldots, a_{m,j})$$

Substituting $\mathsf{T}(\mathbf{v}_j) = a_{1,j}\mathbf{w}_1 + a_{2,j}\mathbf{w}_2 + \cdots + a_{m,j}\mathbf{w}_m$ for each $j = 1, 2, \ldots, n$ into

$$\mathsf{T}(\mathbf{v}) = c_1\mathsf{T}(\mathbf{v}_1) + c_2\mathsf{T}(\mathbf{v}_2) + \cdots + c_n\mathsf{T}(\mathbf{v}_n)$$

and then simplifying we get

$$\mathsf{T}(\mathbf{v}) = \sum_{i=1}^{m}\left(\sum_{j=1}^{n} c_j a_{i,j}\right) w_i$$

Therefore,

$$[\mathsf{T}(\mathbf{v})]_\gamma = \mathbf{A}[\mathbf{v}]_{\mathcal{B}}$$

where $\mathbf{A}$ is the $m \times n$ matrix given by

$$\mathbf{A} = \begin{bmatrix}[\mathsf{T}(\mathbf{v}_1)]_\gamma & [\mathsf{T}(\mathbf{v}_2)]_\gamma & \cdots & [\mathsf{T}(\mathbf{v}_n)]_\gamma\end{bmatrix}$$

The matrix $\mathbf{A}$ is the matrix representation of the linear mapping T in the bases $\mathcal{B}$ and γ.

Example 18.9. Consider the vector space $\mathsf{V} = \mathbb{P}_2[t]$ of polynomial of degree no more than two and let $\mathsf{T} : \mathsf{V} \to \mathsf{V}$ be defined by

$$\mathsf{T}(\mathbf{v}(t)) = 4\mathbf{v}'(t) - 2\mathbf{v}(t)$$

It is straightforward to verify that T is a linear mapping. Let

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{t - 1, 3 + 2t, t^2 + 1\}.$$

(a) Verify that $\mathcal{B}$ is a basis of V.

(b) Find the coordinates of $\mathbf{v}(t) = -t^2 + 3t + 1$ in the basis $\mathcal{B}$.

(c) Find the matrix representation of T in the basis $\mathcal{B}$.

Solution. (a) Suppose that there are scalars c_1, c_2, c_3 such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

Then expanding and then collecting like terms we obtain

$$c_3t^2 + (c_1 + 2c_2)t + (-c_1 + 3c_2 + c_3) = 0$$

Since the above holds for all $t \in \mathbb{R}$ we must have

$$c_3 = 0, \quad c_1 + 2c_2 = 0, \quad -c_1 + 3c_2 + c_3 = 0$$

Solving for c_1, c_2, c_3 we obtain $c_1 = 0, c_2 = 0, c_3 = 0$. Hence, the only linear combination of the vectors in $\mathcal{B}$ that produces the zero vector is the trivial linear combination. This proves by definition that $\mathcal{B}$ is linearly independent. Since we already know that $\dim(\mathbb{P}_2) = 3$ and $\mathcal{B}$ contains 3 vectors, then $\mathcal{B}$ is a basis for $\mathbb{P}_2$

(b) The coordinates of $\mathbf{v}(t) = -t^2 + 3t + 1$ are the unique scalars (c_1, c_2, c_3) such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{v}$$

In this case the linear system is

$$c_3 = -1, \quad c_1 + 2c_2 = 3, \quad -c_1 + 3c_2 + c_3 = 1$$

and solving yields $c_1 = 1, c_2 = 1$, and $c_3 = -1$. Hence,

$$[\mathbf{v}]_{\mathcal{B}} = (1, 1, -1)$$

(c) The matrix representation $\mathbf{A}$ of T is

$$\mathbf{A} = \big[[\mathsf{T}(\mathbf{v}_1)]_{\mathcal{B}} \quad [\mathsf{T}(\mathbf{v}_2)]_{\mathcal{B}} \quad [\mathsf{T}(\mathbf{v}_3)]_{\mathcal{B}}\big]$$

Now we compute directly that

$$\mathsf{T}(\mathbf{v}_1) = -2t + 6, \quad \mathsf{T}(\mathbf{v}_2) = -4t + 2, \quad \mathsf{T}(\mathbf{v}_3) = -2t^2 + 8t - 2$$

And then one computes that

$$[\mathsf{T}(\mathbf{v}_1)]_{\mathcal{B}} = \begin{bmatrix} -18/5 \\ 4/5 \\ 0 \end{bmatrix}, \quad [\mathsf{T}(\mathbf{v}_2)]_{\mathcal{B}} = \begin{bmatrix} -6/5 \\ -2/5 \\ 0 \end{bmatrix}, \quad [\mathsf{T}(\mathbf{v}_3)]_{\mathcal{B}} = \begin{bmatrix} 24/5 \\ 8/5 \\ -2 \end{bmatrix}$$

And therefore

$$\mathbf{A} = \begin{bmatrix} -18/5 & -6/5 & 24/5 \\ \\ 4/5 & -2/5 & 8/5 \\ \\ 0 & 0 & -2 \end{bmatrix}$$

$\square$

After this lecture you should know the following:

- what coordinates are (you need a basis)
- how to find coordinates relative to a basis
- the interpretation of the change-of-coordinates matrix as a mapping that transforms one set of coordinates to another